

Quiz — 1.5.2 Moral and Ethical Issues

Name: _____ Date: _ **Mark:** ____ / 25

OCR H446 · Component 01 · 1.5.2

Section A — Multiple choice (8 marks)

1. A factory replaces assembly-line workers with robots. Which issue does this *most* directly raise? A. Digital divide B. Automation and its impact on the workforce C. Censorship D. Monopolies Your answer: ☐
 2. A bank uses an algorithm trained on historical lending data to approve or reject loan applications automatically. Critics say it unfairly rejects certain ethnic groups. Which issue is this? A. Offshoring B. AI ethics / algorithmic bias C. Environmental impact D. Vendor lock-in Your answer: ☐
 3. Which of the following is a **drawback** of computing for the environment? A. Enables remote working, cutting travel B. Supports smart grids and paperless offices C. E-waste and high energy use of data centres D. Reduces the need for printed documents Your answer: ☐
 4. A government blocks its citizens from accessing certain news websites. Which issue does this raise? A. Censorship and control of the internet B. The digital divide C. Automated decision making D. Layoffs Your answer: ☐
 5. "The gap between people who have access to technology and the skills to use it, and those who do not." This describes: A. Offshoring B. The digital divide C. A monopoly D. Surveillance Your answer: ☐
 6. A company moves its customer-service call centre from the UK to another country to reduce labour costs. This is best described as: A. Automation B. Censorship C. Offshoring D. Open-source licensing Your answer: ☐
 7. Which is a **disadvantage** of one company holding a monopoly over a software market? A. Common standards across products B. Higher prices and vendor lock-in C. Strong technical support D. A polished, well-funded product Your answer: ☐
 8. A self-driving car must "choose" between two unavoidable collisions. Which issue does this scenario best illustrate? A. The digital divide B. The ethics and liability of automated decision making C. Offshoring D. Open-source software Your answer: ☐
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Section B — Short answer (8 marks)

9. Explain **one** benefit and **one** drawback of automated decision making. [2]

10. Explain why an AI system can produce **biased** outcomes. [2]

11. State **two** ways in which the digital divide can disadvantage individuals. [2]

12. Mass surveillance is sometimes defended as crime prevention. State **one** benefit and **one** drawback of large-scale data collection and surveillance. [2]

Section C — Extended response (9 marks)

13. A large supermarket chain plans to replace most of its staffed checkouts with fully automated self-service and "just walk out" till-free technology. Managers say it will cut costs and speed up shopping; staff and unions are concerned about job losses.

Discuss the moral, ethical and social impact of this decision, considering the different stakeholders involved, and reach a justified conclusion. [9]

Answer key

Section A (1 mark each = 8)

1. **B** — Machines doing tasks once done by humans is the automation/workforce issue.
2. **B** — An algorithm reproducing bias from historical data is AI ethics / algorithmic bias.
3. **C** — E-waste and data-centre energy use are environmental drawbacks (A, B, D are benefits).
4. **A** — Blocking access to content is censorship/control of the internet.
5. **B** — This is the definition of the digital divide.
6. **C** — Moving work abroad to cut costs is offshoring.
7. **B** — Monopolies tend to mean higher prices and lock-in (the others are advantages).
8. **B** — The "trolley problem" for autonomous vehicles is about automated decision making and liability.

Section B (8)

9. [2] Benefit (1): fast, consistent, handles huge volumes of data without fatigue/no human error from tiredness. Drawback (1): lacks empathy/human judgement, hard to appeal a decision, or unclear who is liable for a wrong/harmful decision.
10. [2] AI learns patterns from its **training data** (1); if that data reflects historical or societal **bias** (or is unrepresentative), the model **reproduces or amplifies** that bias, and a "black box" model makes it hard to detect or explain (1).

11. **[2]** Any two, 1 each: cannot access online-only services (banking, government forms); disadvantaged in education/jobs that require digital skills; excluded from cheaper online prices/deals; isolated socially; particularly affects poor, rural, elderly or disabled people — widening inequality.
12. **[2]** Benefit (1): helps prevent/detect crime or terrorism, supports public health, enables personalised services. Drawback (1): loss of privacy, profiling, risk of data breaches/misuse, or a "chilling effect" on free expression.

Section C — indicative content (9)

This is a levels-of-response "**discuss**" question; mark the **whole answer**, not point-by-point. Expect a balanced discussion across multiple stakeholders with a justified conclusion.

Benefits (FOR): - *Supermarket/business*: lower wage costs, faster throughput, fewer queues, more competitive prices. - *Customers*: quicker, sometimes 24/7 service; potentially lower prices. - *Some new roles*: technicians to maintain/repair the systems.

Drawbacks (AGAINST): - *Employees*: job losses / layoffs, especially low-skilled checkout staff; deskilling; regional unemployment. - *Society*: loss of human contact (a problem for elderly/vulnerable shoppers); widening of the digital divide for those uncomfortable with the technology; data/privacy concerns with "just walk out" tracking and surveillance. - *Ethical*: is it right to prioritise profit over jobs; corporate responsibility to retrain staff.

Stakeholders to name (aim 3–4+): the business/shareholders · employees · customers (incl. elderly/vulnerable) · wider society/local community.

Justified conclusion: must take a clear position, ideally conditional — e.g. "*automation is justified provided staff are retrained or redeployed and some staffed tills remain for those who need assistance.*"

Levels of response (9 marks): - **Level 1 (1–3):** Fragmented, one-sided list (e.g. only job losses or only cost savings). Little development, weak/no use of terminology, no conclusion. - **Level 2 (4–6):** Some discussion of both sides but mostly descriptive or one-sided; limited stakeholders; conclusion weak, unsupported or absent. - **Level 3 (7–9):** Balanced and well-developed, several stakeholders, **both** benefits and drawbacks with accurate terminology and sustained reasoning, leading to a **justified conclusion** that takes a reasoned (often conditional) position.

Total: 8 + 8 + 9 = 25